

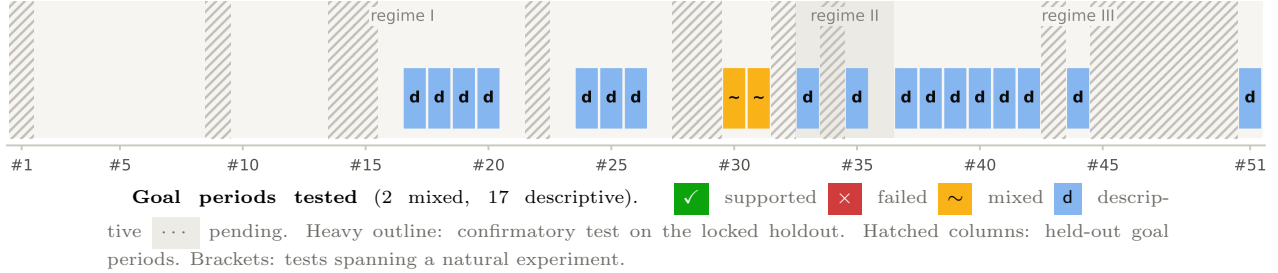
H27: Critical slowing down warns of herding waves hours ahead

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 2/5

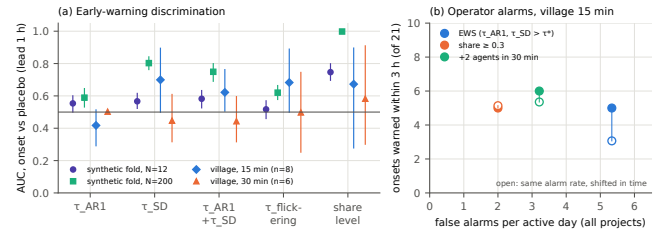
Bottom line. Critical slowing down does not warn of herding in the AI Village: pile-ons are fast, often announced by a link, and too few agents are watched for slowing down to show.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

A pre-registered rule on H11's project labels defined herding onsets: a project's share of labeled agents jumps from ≤ 0.25 to ≥ 0.5 within an hour and stays there. It found 21 onsets in 14 non-holdout weeks at 15-min resolution, mostly in shared-artifact weeks (#18, #19, #30, #31, #33, #38). Early-warning trends (Kendall τ of the rolling lag-1 autocorrelation and SD over the previous 6 h) were compared with matched placebo segments. Only 8 onsets were evaluable. The composite separated them weakly at 15 min (AUC 0.62 [0.51, 0.76]) and not at 30 min (0.44). Autocorrelation never rose (AUC 0.42), so this is not critical slowing down. The frozen alarm warned of 5 of 21 onsets at 5.3 false alarms per active day, no better than a time-shifted alarm ($p = 0.07$) or a naive share alarm. Four onsets came out of nowhere. Post hoc, 11 of 21 followed a chat link to the project within 30 min ($p = 0.0005$). Holdout not run.



Observables. (a) AUC of each early-warning trend, onset vs placebo segments, at 1 h lead. Series: a synthetic fold at village size (purple), a clean $N = 200$ fold (green), and the village at 15 min (blue, 8 onsets) and 30 min (orange, 6). Autocorrelation never discriminates; variance does only at 15 min, with wide CIs. (b) Onsets warned within 3 h against false alarms per active day. Open circles: the same alarms shifted in time. No alarm beats its shifted copy.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator this is a useful negative. Generic early-warning indicators (rising autocorrelation and variance, popular in ecology and finance) do not forecast pile-ons in a 10–15-agent LLM swarm logged at 15-min resolution. Most pile-ons are fast pushes: a project nobody had touched, or a link posted in chat, then a majority within the hour. The synthetic runs show it would not help even if the physics were right: at $N \approx 12$ a true fold is barely detectable. So don't build a critical-slowing-down dashboard for swarms this size. If you want warning, watch announcements (links; H28 is testing this) or just watch shares, and accept about two false alarms a day. The transferable lesson: test alarm timing against a rate-matched shift, or every indicator looks predictive.

PHYSICAL MODEL

Models 10 (kinetic mean-field Potts) and 01 (mean-field susceptibility). σ_i : agent i 's project in a window (0 = rare projects, uncoupled). $k_a^{(-i)}$: other agents on a ; $h_a(t)$: a slowly drifting popularity field; βJ : uniform herding coupling; γ : reconsideration rate; x_a : share on a . Near a fold of x_A the leading eigenvalue $\lambda \rightarrow 0$, so the lag-1 autocorrelation ρ_1 and the variance rise (critical slowing down). Rivals (field steps, noise-induced flips) give no warning.

$$P(\sigma_i \rightarrow a) \propto e^{h_a(t) + \beta J k_a^{(-i)} / (N-1)} \quad (a \geq 1), \quad \propto 1 \quad (a = 0)$$

$$\dot{x}_a = \gamma [\text{softmax}_a(h(t) + \beta J x) - x_a]$$

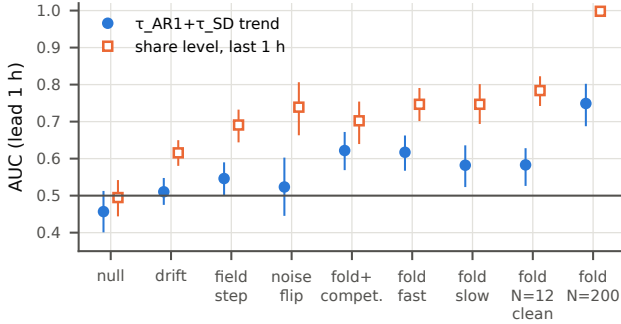
$$\rho_1 = e^{-\gamma |\lambda| \Delta t} \rightarrow 1, \quad \text{Var } x_A \propto 1/(N|\lambda|) \quad (\lambda \rightarrow 0)$$

$$\text{alarm: } \tau_K(\hat{\rho}_1) > \tau^* \wedge \tau_K(\hat{\sigma}) > \tau^*, \quad \tau^* = 0.54$$

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RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P0: 10–25 onsets	21 in 7/14 weeks; 8 evaluable	yes
P1: $AUC \geq 0.70$	0.62 [0.51, 0.76]; 30 min: 0.44	mixed
P2: $SD > AR1$, mean rise	0.70 vs 0.42; binomial 0.69	partly
P3: no operator value	5/21 hits, 5.3 false/day, $p = 0.07$	yes
P4: synthetic	$N=200$: 0.75; $N=12$: 0.58	2 of 4
P5: lead ≤ 1 h	1.25 h (inflated)	moot
P6: concentration	2 evaluable onsets	untestable
Post hoc: link first	11/21 vs 2.7 expected; $p = 0.0005$	to confirm
P3's p compares the alarm with the same alarms shifted in time. P4 missed on $N=200$ (0.8 predicted) and on drift raising false alarms (it did not).		



Second figure. Synthetic kinetic Potts at village sampling, 200 runs per scenario. Filled: the composite trend; open: the share level in the last hour. The nulls, field steps and noise-induced flips sit near 0.5. A true fold at $N = 12$ reaches only 0.58–0.65; a clean $N = 200$ fold, 0.75. The share level beats the trends everywhere.

GOAL PERIODS IN DETAIL

G17 d descriptive (30-min arm; no evaluable onset)
G18 d descriptive (1 evaluable onset, percentile 0.46)
G19 d descriptive (1 evaluable onset, percentile 0.59)
G20 d descriptive (30-min arm; no evaluable onset)
G24 d descriptive (30-min arm; no evaluable onset)
G25 d descriptive (30-min arm; 1 evaluable onset, percentile 0.82)
G26 d descriptive (30-min arm; no evaluable onset)
G30 ~ mixed (AUC 0.57, 2/4 hit)
G31 ~ mixed (AUC 0.91, 1/4 hit)
G33 d descriptive (1 evaluable onset, no placebos)
G35 d descriptive (no evaluable onset)
G37 d descriptive (no evaluable onset)
G38 d descriptive (1 evaluable onset, percentile 0.86)
G39 d descriptive (no evaluable onset)
G40 d descriptive (no evaluable onset)
G41 d descriptive (no evaluable onset)
+3 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	0	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

21 onsets, 8 evaluable: the per-week AUCs rest on 1–2 onsets and the CIs are wide. Many indicators, leads, window lengths and two onset rules were computed; only the 15-min composite at 1 h was primary. Labels are artifact mentions (attention, not work) and are sparse in regime I. The step rule misses slow rises (synthetic $N = 200$); the slow-rise variant gives 0.73 at 15 min but 0.50 at 30 min. The link association is post hoc and partly mechanical, since a link also labels its poster. #51 supplies most of the false alarms. Leads count from the earliest alarm in the horizon, which flatters noisy alarms.

ROUND 2

- **Essence.** Can an operator see a pile-on coming early enough to act, and from what signal?
- **H27-R1** Model onsets as announcement-triggered cascades: hazard of a majority pile-on after a link post, by poster, room size and the project's prior share; joint with H28.
- **H27-R2** Use a finer clock: per-turn project states from actions (minutes, not 15-min windows) so 6 h of history is not needed and the first 1.5 days become evaluable.
- **H27-R3** Test the one regime where the theory says warnings can work: long, large swarms with slowly drifting incentives (#51 roles, if any shared project ever rises), with the slow-rise onset rule.

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